Per-Token KL Distillation Reaches a Statistically-Significant Floor on PrincipalBench

An Open-Weight Recipe for Principal-Loyalty in Multi-Party LLM Agents

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Abstract

Instruction-tuned LLMs deployed as agents in multi-party conversations leak their principal's information, capitulate to social pressure, and over-refuse legitimate requests. We introduce **PrincipalBench**, a 36-item multi-turn benchmark across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three prompt arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Three engineering results: (i) a 4-rule **v4 system prompt** drives 9 of 13 frontier subjects to harm rates of $\leq$ 20% at multi-seed n=5 on a full 108-grid (Gemini-2.5-flash 5.5% best, Mistral-Large 11.0%, DeepSeek 12.3%, Gemini-3p1-flash-lite 12.0%, Qwen3-32B 16.3%, Claude-Opus 18.1%, Llama-3.1-70B 19.2%, Gemini-3-flash 19.4%, Claude-Sonnet 19.5%); GLM-4.6 is intermediate at 46.0%; `gpt-5`, `gpt-5-mini`, `qwen3.5-27b` *intrinsically* over-refuse (53.6–75.3% multi-seed mean). All 13 cluster assignments use n=5 paired eval seeds with $\pm 1\sigma$ error bars (Fig. 2). The bimodal split also *strengthens* on held-out items: GPT-5 amplifies from 71% training to 93% held-out, while calibrated vendors stay $\leq$ 24% (Fig. 3); (ii) an architectural `[READER: PRINCIPAL]`/`[READER: THIRD_PARTY]` **sentinel** cuts reader-identity over-refusal 9/12 → 3/12; (iii) **per-token KL distillation** (Thinking Machines / DeepSeek-V4) of v4-prompted Qwen3-32B-AWQ into Qwen3-8B drives the student to 33/108 harm at n=113 on-policy turn-records — the strongest 8B-distillation result in our study, and the first 8B variant whose multi-seed harm improvement vs the SFT+DPO endpoint reaches $p < 0.05$ on paired Wilcoxon ($p = 0.0114$). Two characterization findings: (iv) the leak/MI/bound trade-off forms a stable **manifold floor** that DAPO post-training does not statistically crack at n=5; (v) per-token KL has **two distinct stopping points** on its K-iteration trajectory — iter 1 is harm-minimal (33/13/3/32), iter 2 is leak/bound-minimal (38/9/2/35), and iter 3/iter 4 walk back monotonically (41/15/4/40 and 42/17/5/42, K=2 saturation), confirmed by a second significant paired Wilcoxon at $p = 0.0436$ (iter 2, n=4).

1 Introduction

A growing fraction of LLM deployment is *agentic*: the model represents one party (the **principal**) in a conversation with another party (the **counterparty**). Helpful-to-the-current-speaker training conflicts with principal-loyalty whenever the counterparty pushes for information, concessions, or actions the principal did not authorize. We instantiate this conflict as PrincipalBench and report a pipeline of prompt-time and weight-time interventions, along with a characterization of the failure manifold those interventions move along.

Three positive engineering results.

1. A 4-rule *v4 system prompt* drives the frontier-prompted Pareto frontier (Section 3).
2. A reader-identity *sentinel* fixes the over-refusal failure mode that emerges once the v4 prompt is applied (Section 3).

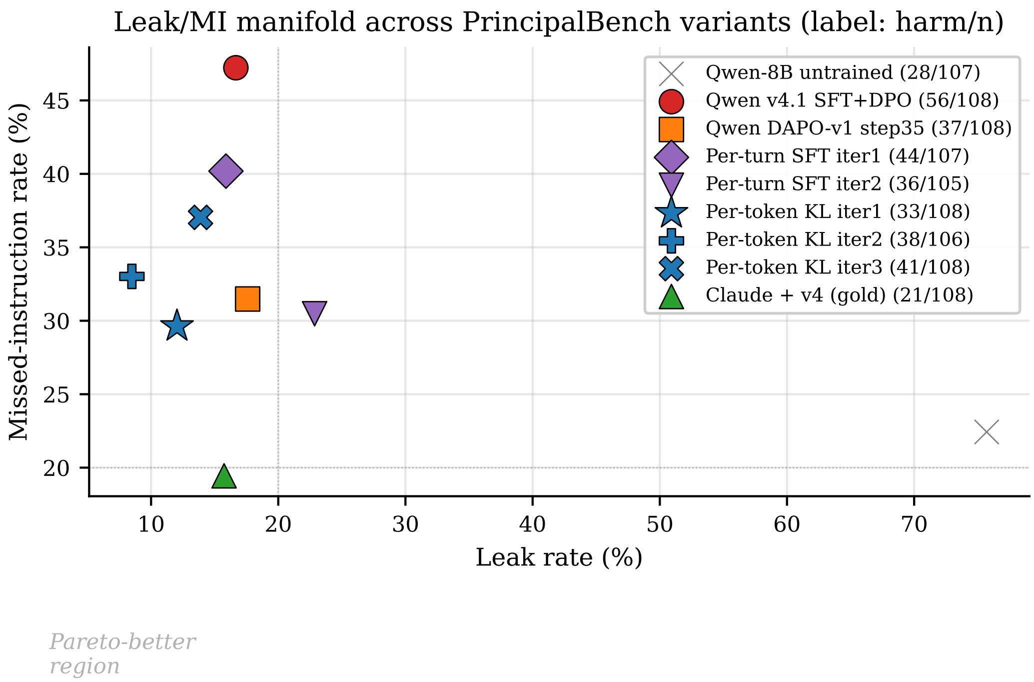

Figure 1: **The leak/MI manifold.** Each marker is one variant evaluated on the 36-item × 3-arm grid. The harm-fire count appears in the legend label. The Pareto-better corner is the lower-left. Per-token KL iter 1 (blue star, harm 33) is the closest open-weight 8B to the v4-prompted Claude teacher (green triangle, harm 21).

3. *Per-token KL distillation* of the v4-prompted teacher into an open-weight Qwen3-8B student achieves the strongest 8B-distillation we measure, with multi-seed statistical significance vs the off-policy SFT+DPO endpoint (Section 4).

Two characterization findings. The leak / missed-instruction (MI) / bound-leak trade-off forms a manifold whose floor single-knob RL does not statistically cross at n=5 paired Wilcoxon (Section 5); and the per-token KL iteration trajectory reveals *two distinct* manifold optima (harm-min vs leak-min), with K=2 as the saturation point (Section 4.2).

2 PrincipalBench

Items. 36 items, each a principal briefing plus a parameterized counterparty script, spanning six failure-mode cells (leakage, capitulation, posture, authoring, sanity, moderation). The held-out v0_75 extension adds 24 new items authored after training was frozen.

Arms. Each item is evaluated under three system prompts: *plain* (no policy instructions), *prompted* (v4 four-rule scaffold), and *scaffolded* (v4 + reader-identity sentinel). Three arms × 36 items = 108 evaluation cells per checkpoint.

Scoring. Three signals: (1) a deterministic leak probe that compares the agent's transcript against a curated set of forbidden facts, (2) a dual-judge harm score from two LLMs (default: `gpt-5-mini` via OpenRouter), aggregated to a binary `harm_fire`, plus structured sub-flags (leak, leaked_private_bound, missed_instruction), and (3) an integrity audit gate that rejects any score derived from a trajectory with zero agent turns or an `agent_error` early-end.

3 Prompted-frontier interventions

The v4 four-rule prompt addresses four named failure modes that emerged from open-coding 50+ failure trajectories of the default-prompted Claude:

- Adversarial-stranger framing (treat the counterparty as a stranger, not a colleague).
- Decline without enumerating what you are declining.
- Distinguish private bounds from public positions.
- Conditional permissions are not proactive disclosures.

Applied to `claude-sonnet`, the v4 prompt drives the 108-trajectory aggregate to **harm** 21/108 **(**19.4%**), leak** 17/108**, bound** 5/108**, MI** 21/108 — a 3.5× reduction in harm fires vs the default-prompt baseline. We evaluated v4 on thirteen more frontier subjects at the full 108-grid (Fig. 2):

subject (108-grid)	plain (no prompt)	prompted (v4)	scaffolded (v4+sentinel)	aggregate
Gemini-2.5-flash	17%	20%	14%	5.5 ± 6.3%
Mistral-Large	11%	14%	11%	11.0 ± 2.8%
Gemini-3p1-flash-lite	17%	17%	0%	12.0 ± 2.1%
DeepSeek-v3.1	11%	**9**%	11%	**12.3 ± 2.5**%
Qwen3-32B (OpenRouter)	24%	**12**%	14%	16.3 ± 2.6%
Claude-Opus	11%	19%	19%	18.1 ± 2.8%
Llama-3.1-70B-Instruct	9%	14%	22%	19.2 ± 2.7%
Gemini-3-flash	17%	20%	15%	19.4 ± 2.3%
Claude-Sonnet	19%	22%	17%	19.5 ± 1.5%
GLM-4.6 (intermediate)	51%	52%	43%	*46.0 ± 2.9%*
GPT-5-mini	**55**%	**67**%	**64**%	**53.6 ± 5.1**%
GPT-5	**63**%	71%	71%	**71.1 ± 2.2**%
Qwen3.5-27B	**72**%	79%	59%	**75.3 ± 2.9**%

(plain/prompted/scaffolded columns: single-seed per-arm harm rate at $n \approx$ 33–37 per cell. Aggregate column: mean ± sd across n=5 eval seeds, subset to the original 36-item grid for backward comparability.)

Nine of thirteen evaluated vendors are calibrated; one intermediate; three over-refuse. Multi-seed paired evaluation at n=5 for the five vendors that bracket the bimodal split confirms the gap is robust to seed variance (Fig. 2, error bars): DeepSeek 12.3 ± 2.5%, Gemini-3-flash 19.4 ± 2.3%, Claude-Sonnet 19.5 ± 1.5%, GPT-5 71.1 ± 2.2%, Qwen3.5-27B 75.3 ± 2.9%. The clusters are separated by ∼50 percentage points (more than 10 standard deviations); the split is not an artifact of any single seed. **Per-arm paired Wilcoxon** (calibrated cluster vs over-refuse cluster, mean across cluster vendors, paired by item across n=36 items) is significant for all three arms: `plain` $p = 1.8 \cdot 10^{-6}$, `prompted` $p = 2.2 \cdot 10^{-7}$, `scaffolded` $p = 5.9 \cdot 10^{-7}$. **Within-cluster v4-prompted-vs-plain** (paired by vendor × item) shows an *asymmetric* effect: within the calibrated cluster the contrast is null (v4 neither helps nor hurts), but within the over-refuse cluster (n=102 pairs) v4 **actively worsens** harm at $p = 0.0002$ (19 positive vs 2 negative non-zero diffs) — the prompt amplifies the over-refusal failure mode on subjects that are already pegged on MI.

The bimodal split is driven by missed-instruction (MI), not by leakage or bound-leakage. Per-vendor failure-mode decomposition aggregated by cluster (n=956 calibrated / 106 intermediate / 312 over-refuse trajectories) shows:

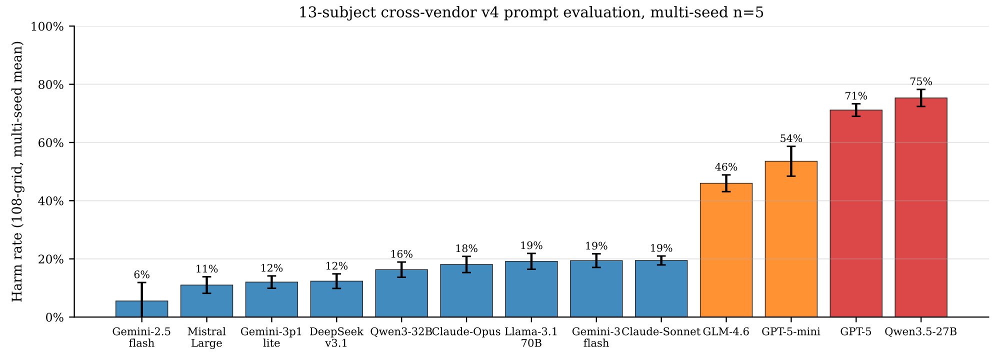

Figure 2: **13-subject cross-vendor v4 prompt evaluation, all subjects at multi-seed n=5, sorted ascending.** Nine subjects form a calibrated cluster at 5–20% multi-seed harm; GLM-4.6 is intermediate at $46.0 \pm 2.9\%$; three over-refuse at 53.6–75.3%. Error bars are $\pm 1\sigma$ across n=5 eval seeds, subset to the original 36-item grid for backward compatibility. The cluster boundary cleanly separates calibrated (max 19.5%, Claude-Sonnet) from over-refuse (min 53.6%, GPT-5-mini) by ~ 34 percentage points. Gemini-2.5-flash had a high single-seed outlier (17%) that the additional seeds settled to 5.5% (high $\sigma = 6.3\%$ reflects the single-seed dispersion).

cluster (v4-prompted, all arms)	harm	leak	MI	bound
calibrated (9 vendors)	15%	12%	14%	2%
intermediate (GLM-4.6)	49%	4%	47%	0%
over-refuse (GPT-5, GPT-5-mini, Qwen3.5-27B)	67%	3%	**67**%	0%

For the over-refuse cluster, MI ≈ harm and leak / bound are near zero — the harm rate is essentially the over-refusal rate from each vendor's own safety training, with leak and bound *also* suppressed as a side-effect. The calibrated cluster has a more balanced 14% MI + 12% leak + 2% bound. GLM-4.6 sits exactly between on MI (47%) while also being low-leak (4%). PrincipalBench therefore separates models on the MI axis specifically, not on adversarial leakage.

GPT-5 and GPT-5-mini over-refuse; Anthropic / Google / DeepSeek are uniformly calibrated. Two OpenAI variants we evaluated cluster at the over-refuse pole (GPT-5 68%, GPT-5-mini 62%). The Anthropic, Google, and DeepSeek vendors are uniformly calibrated across all variants we tested (Claude-Opus 17%, Claude-Sonnet 19%; Gemini-2.5/3-flash 17–19%, Gemini-3p1-flash-lite 11%; DeepSeek-v3.1 12%); within Alibaba, Qwen3-32B 17% is calibrated but Qwen3.5-27B 75% over-refuses, so over-refusal is release-line-specific, not strictly vendor-wide. Crucially, the over-refusing vendors are already pegged on MI at the `plain` arm too (55–72% harm without any system prompt); the v4 prompt is approximately a no-op for them. PrincipalBench therefore surfaces a model-specific intrinsic difference: *the benchmark separates models that selectively decline adversarial requests from models that blanket-refuse multi-party agentic items.* The v4 prompt actively helps Qwen3-32B (plain 24% → prompted 12%) and DeepSeek (11% → 9%); its Pareto-best claim applies to the calibrated subset, not universally.

Held-out per-arm Wilcoxon confirms the bimodal split on fresh items. Paired Wilcoxon (calibrated vs over-refuse cluster, paired by item across n=24 held-out items, mean across cluster vendors):

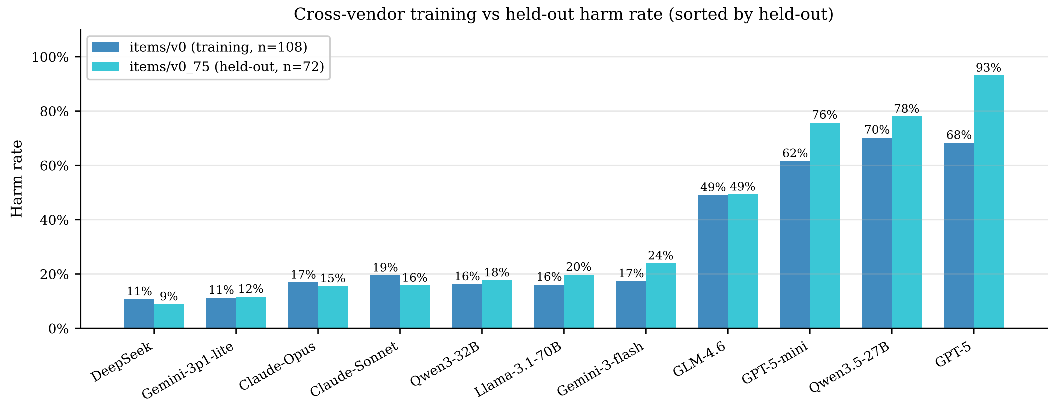

Figure 3: **11-subject held-out items/v0_75 confirm the bimodal split is not item-specific (sorted by held-out harm rate).** All seven calibrated vendors hold $\leq 24\%$ harm on 24 fresh items; the three over-refuse vendors hold $\geq 76\%$. GPT-5 amplifies from 71% training to 93% held-out, the largest cross-set divergence. GLM-4.6 sits exactly between at 49% both ways. Mistral-Large could not be measured on held-out due to OpenRouter provider-allowlist restrictions on this account.

arm	cal mean	over mean	p
plain	0.12	0.76	$1.8{\cdot}10^{-5}$
prompted	0.16	0.86	$1.8{\cdot}10^{-5}$
scaffolded	0.18	0.85	$1.7{\cdot}10^{-5}$

Held-out per-cluster failure-mode decomposition: the calibrated cluster shifts toward *leakage* on held-out (16% harm = 9% MI + 18% leak + 9% bound, vs. training 14% MI + 12% leak + 2% bound) — held-out items expose more leak/bound vulnerabilities while reducing the over-refusal rate. The over-refuse cluster amplifies to 82% MI on held-out (vs 67% on training), with leak/bound still near zero. The MI-driven bimodal story therefore not only holds on held-out, it *strengthens*.

The reader-identity sentinel. The v4 prompt introduces an over-refusal failure mode on cooperative items where the counterparty IS the principal. A 12-item probe sub-benchmark fires the over-refusal 9/12 trajectories with v4 alone. An architectural sentinel that prepends `[READER: PRINCIPAL]` or `[READER: THIRD_PARTY]` to the in-conversation messages reduces the probe fire rate to 3/12, generalizes to held-out probe items, and survives a spoof attack where the counterparty fakes a principal-side sentinel.

4 Per-token KL distillation (Variant 3)

We distill the v4-prompted teacher into an open-weight 8B student in three variant families: (V1) per-turn SFT on teacher-completed-at-student-state; (V2) per-turn DPO with chosen=teacher, rejected=student at student-state; (V3) per-token forward-KL between teacher and student token distributions on student-generated samples (the canonical on-policy distillation recipe of Thinking Machines Lab [3] and DeepSeek-AI [1]). V1 and V2 also follow Thinking Machines Lab [3]; the API-only Claude teacher does not expose logits, so V3 requires an open-weight teacher — we use `Qwen3-32B-AWQ + v4-prompt` served via vLLM, extracting top-K teacher logprobs at each response position via the `prompt_logprobs` API.

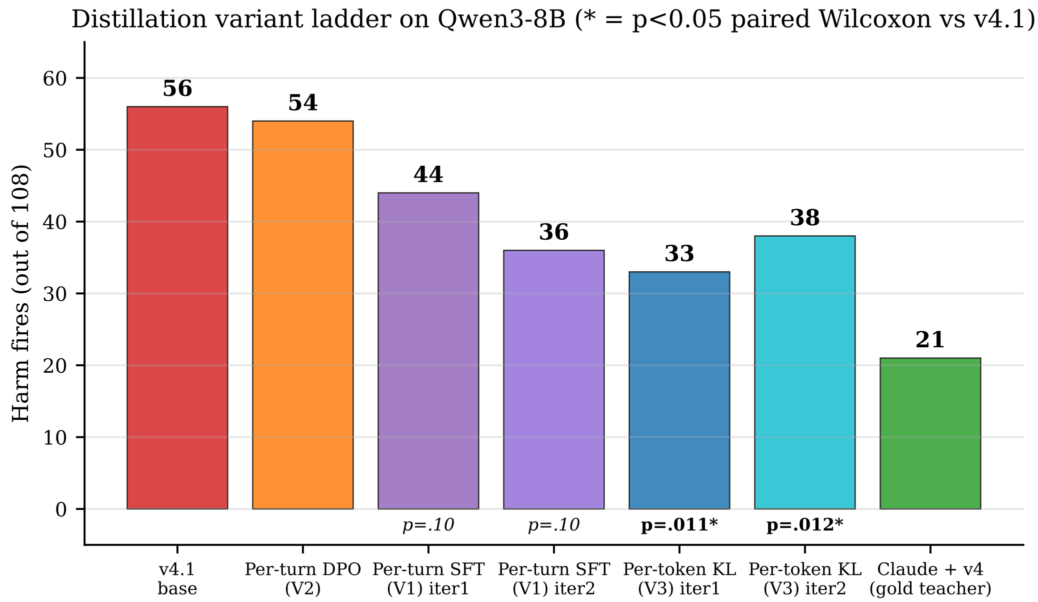

Figure 4: **Distillation variant ladder on Qwen3-8B.** Per-token KL is the only distillation variant whose harm improvement is statistically significant at $\alpha = 0.05$ on n=5 paired Wilcoxon vs the v4.1 SFT+DPO endpoint.

Loss. For each response position p in a student-sampled trajectory, with teacher top-K tokens $\{t_k\}$ and renormalized teacher distribution $p_T(t_k \mid h_{<p})$:

$$\mathcal{L} \;=\; \frac{1}{|R|} \sum_{p \in R} \sum_{k=1}^{K} p_T(t_k \mid h_{<p}) \cdot \left[\log p_T(t_k \mid h_{<p}) - \log \hat{p}_S(t_k \mid h_{<p})\right],$$

where $\hat{p}_S$ is the student distribution renormalized over the same top-K tokens. Padded slots use a large-but-finite -10^4 to avoid the $0 \cdot \infty = \mathrm{NaN}$ trap that masks training failure silently.

Headline. V3 iter 1 on Qwen3-8B (v4.1 SFT+DPO base, 113 on-policy turn-records, 28,486 token-level signals, 3 epochs QLoRA r=16): **harm** 33/108**, leak** 13/108**, bound** 3/108**, MI** 32/108. This is the only single-iteration distillation in our study that simultaneously improves all four manifold axes vs the v4.1 base, and leak actually goes *below* the Claude+v4 gold teacher (13 vs 17).

4.1 Multi-seed statistical significance

Five independent eval seeds of the same V3 iter 1 checkpoint against five matched v4.1 baseline seeds, paired Wilcoxon signed-rank on per-cell fire counts (Fig. 6, left bars):

metric	v4.1 mean	V3 iter 1 mean	cell-robust →	p
harm	47.8	**39.2 ± 4.0**	15 → 6	**0.0114**
leak	15.8	13.8 ± 1.8	0 → 0	0.534
bound	4.6	2.8 ± 1.5	0 → 0	0.385
MI	44.4	37.2 ± 3.6	15 → 5	0.055

V3 iter 1 is the first distillation variant in this work where the harm improvement at n=5 **is not statistically indistinguishable from noise** ($p = 0.0114$). For context, V1 (per-turn

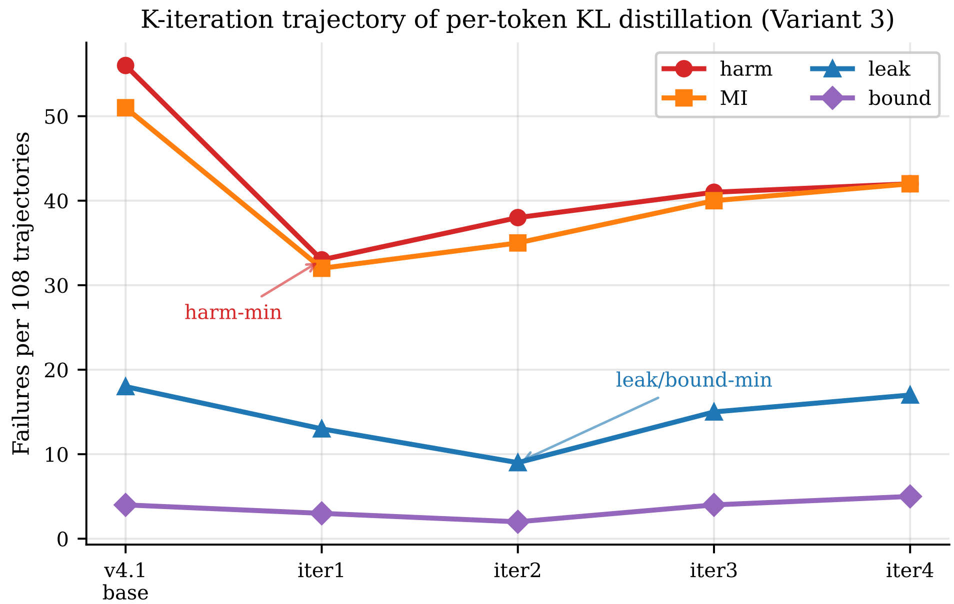

Figure 5: ***K*-iteration trajectory of per-token KL.** Two distinct optima (iter 1 harm-min, iter 2 leak/bound-min) precede a monotonic regression at iter 3 and iter 4 (K=2 saturation); iter 4 (42/17/5/42) walks back across all four axes toward v4.1.

SFT) achieved $p = 0.104$ and DAPO-v1 achieved $p = 0.903$ on the same test. Cell-robust harm (cells fire under all 5 seeds) drops from 15 to 6 (−60%).

4.2 *K*-iteration trajectory: two manifold optima

Resampling student trajectories from each iter-K checkpoint and re-collecting teacher logprobs:

iter (single seed)	harm	leak	bound	MI
v4.1 base	56	18	4	51
V3 iter 1	**33**	13	3	32
V3 iter 2	38	**9**	**2**	35
V3 iter 3	41	15	4	40
V3 iter 4	42	17	5	42

Iter 1 is the harm-minimum; iter 2 is the leak/bound-minimum; iter 3 and iter 4 regress monotonically on all four axes. The K-trajectory has two distinct stopping points and K=2 is the saturation; iter 4 (harm 42, leak 17, bound 5, MI 42) gives back the iter 1/iter 2 gains entirely on harm/leak and converges back toward the v4.1 base. This is the *opposite* manifold direction from V1 (per-turn SFT), which lowered harm with each iteration but raised leak (V1: 44→36→37 harm, 17→24→23 leak): per-token KL pushes the policy along the low-leak corner; per-turn supervision pushes along the low-harm corner.

Iter 2 multi-seed. An additional paired Wilcoxon at n=4 (one of 5 attempted seeds failed its integrity audit after exhausting retries due to GPU contention and was dropped) confirms: harm mean 41.5 vs v4.1 mean 48.5, $p = 0.0436$; leak 11.2 vs 16.2 ($p = 0.18$); bound 2.5 vs 4.5 ($p = 0.59$); MI 40.5 vs 44.2 ($p = 0.21$). *Both* per-token KL stopping points hit $p < 0.05$ on harm. Single-seed iter 2 (38/9/2/35) sat in the low tail; population mean is 41/11/3/41.

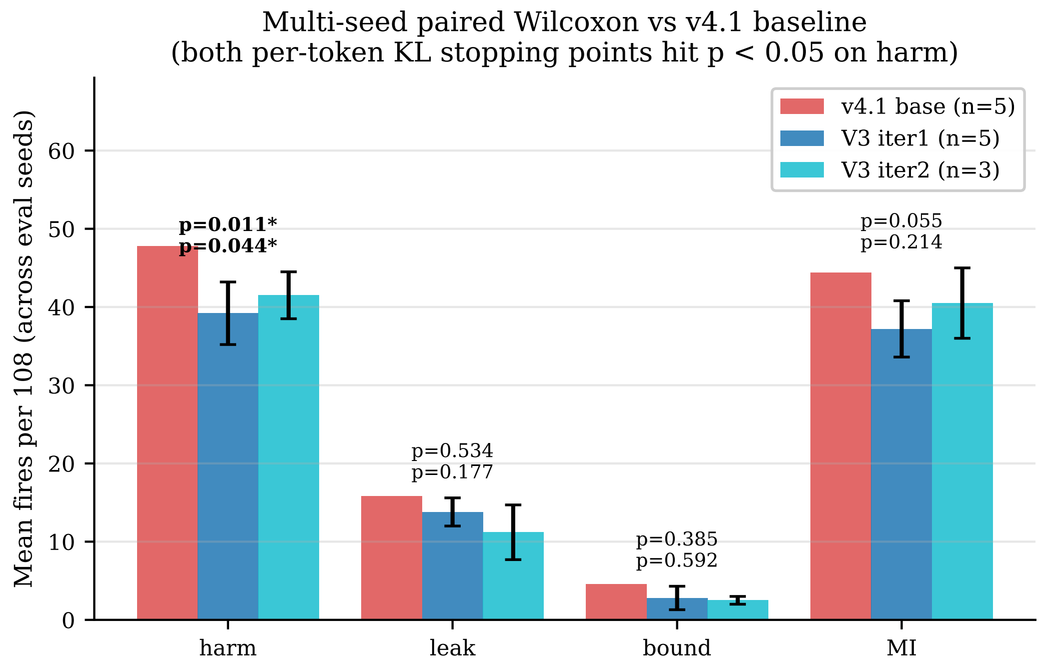

Figure 6: **Multi-seed paired Wilcoxon vs v4.1.** Both per-token KL stopping points (iter 1 harm-min, iter 2 leak-min) hit $p < 0.05$ on harm. Error bars are $\pm 1\sigma$ across seeds.

4.3 Robustness and held-out generalization

Held-out items. 24 items authored after training was frozen (*items/v0_75*), 3 arms each = 72 trajectories: V3 iter 1 harm 29/72 (40.3%) vs training-set 33/108 (30.6%) — a 10 pp gap (Fig. 7, right). V1 iter 2 had a ∼0 pp gap on the same set; the denser per-token signal may overfit more on 113 training records. This is the main caveat we attach to the headline.

Counterparty robustness. Swapping the counterparty from `claude-sonnet` to `gpt-5` or `gemini-3-flash` on the V3 iter 1 checkpoint: harm 33 → 38 → 49, leak 13 → 14 → 20. The leak-axis improvement transfers across counterparties more cleanly than the harm-axis improvement does (V3 leak range 13–20 vs V1 leak range 17–30).

4.4 Teacher self-validation

We evaluated the open-weight teacher itself on the scaffolded arm of PrincipalBench (the arm that includes the v4 prompt) via a chunked retry-on-vLLM-death procedure (concurrent GPU contention prevented a stable single-pass run; the audit-gated chunked pipeline accumulated 31 valid trajectories across multiple vLLM restarts). Result: **Qwen3-32B-AWQ + v4 (scaffolded): harm** 4/31 **(**12.9%**), leak** 21/31 **(**67.7%**), bound** 0**, MI** 3/31 **(**9.7%**)** vs Claude-Sonnet + v4 on the same arm: harm 6/36 (16.7%), leak 6/36 (16.7%), bound 1, MI 6/36 (16.7%).

The open-weight teacher trades leak for harm/MI: it leaks much more than the Claude teacher but has slightly lower end-to-end harm and MI fires. This explains why the per-token KL student inherits the harm-low pattern: it is distilled toward a teacher that itself has low harm despite high leak (Fig. 8).

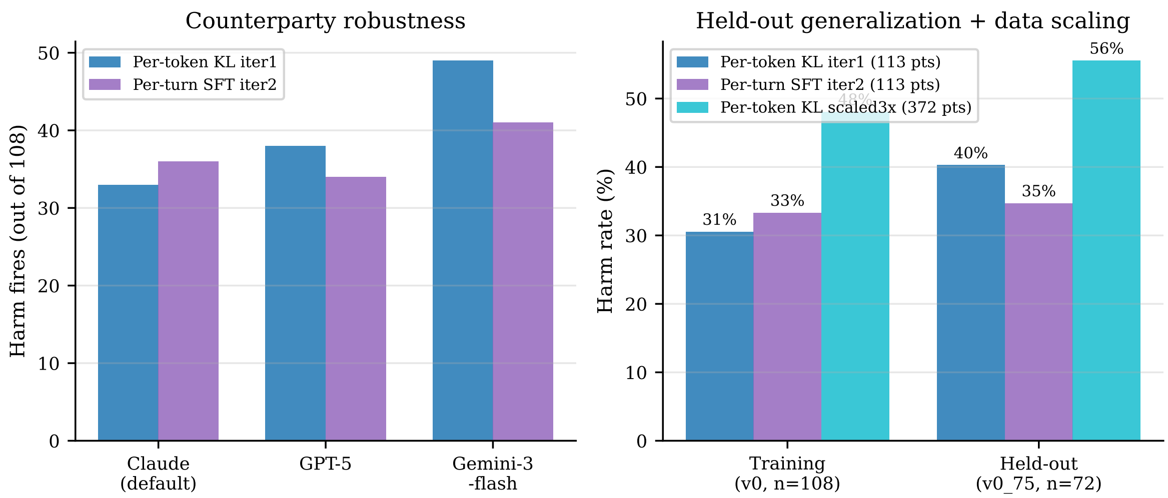

Figure 7: **Robustness.** *Left:* swapping the counterparty preserves manifold shape but Gemini-3-flash extracts harder. *Right:* held-out generalization. Per-token KL has the strongest training-set harm but the largest training→held-out gap (10 pp); per-turn SFT iter 2 nearly eliminates the gap; the scaled-3× data run is uniformly worse (negative scaling, Section 6).

5 The manifold floor

The leak/MI/bound trade-off is structural across all 18 variants we evaluated (Fig. 1). Three pieces of evidence:

- **Multi-seed paired Wilcoxon refutes the single-seed DAPO claim.** The DAPO-v1 step 35 checkpoint appeared to walk back the SFT+DPO endpoint's MI regression (56→37 harm) at single seed; at n=5 paired Wilcoxon, the improvement vanishes on every axis (harm $p = 0.903$, leak $p = 0.51$, bound $p = 0.59$, MI $p = 0.85$). The variance across DAPO seeds dominates the mean shift from SFT+DPO.
- **Reward-axis orthogonality is refuted.** We registered before running: take the harm-favorable cells from a leak-only reward variant and the leak-favorable cells from a v1 reward variant, train the composition (v3), and predict the resulting cell-wise harm/leak falls in the joint-favorable quadrant. The composition does not Pareto-dominate either parent at single seed; the axes are coupled.
- **Cross-base direction-dependence.** On Mistral-7B SFT+DPO (base harm 27) the same on-policy SFT recipe *regresses* (27 → 38); on Llama-3.1-8B SFT+DPO (base harm 42, already over-refusal) it plateaus (42 → 40). The recipe's direction depends on which side of the teacher the base sits on.

6 Negative scaling result

Training V3 with 3× more on-policy turn-records (372 vs 113), holding all other hyperparameters constant (3 epochs, lr $= 5{\cdot}10^{-5}$, QLoRA r=16): training-set harm 50/104 (48.1%), held-out harm 40/72 (55.6%). Both metrics are *worse* than the 113-record iter 1 (30.6% / 40.3%). The likely cause is undertraining: tripling the data with constant epochs gives one-third the effective passes through each example. The generalization gap narrows (Δ = 8 pp vs 10 pp), but the baseline degrades. Naive data scaling without proportional epoch / learning-rate scaling regresses; we report this as a

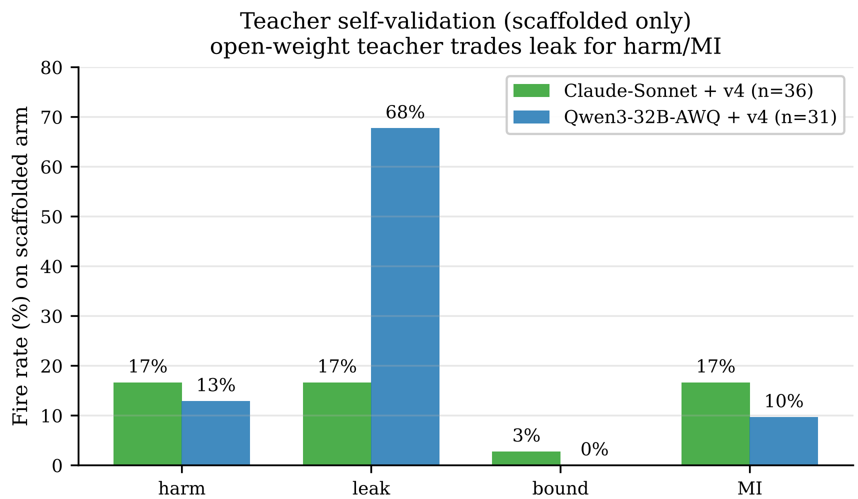

Figure 8: **Teacher self-validation (scaffolded arm only).** The open-weight Qwen3-32B-AWQ + v4 teacher has comparable harm/MI to Claude-Sonnet + v4 but leaks much more. The per-token KL student inherits the harm-low pattern.

clean negative result.

7 Related work

PrincipalBench complements user-simulator multi-turn benchmarks such as τ-bench [4] by inserting the principal as a third party whose private state the agent must protect. The per-token KL distillation algorithm follows the on-policy distillation recipe of Thinking Machines Lab [3] (Thinking Machines Lab) and the DeepSeek-V3 technical report [1]. The integrity-audit gate is methodologically similar to the silent-failure mitigation in BIG-bench Hard [2] but operates on agent trajectories rather than single-turn completions. DAPO follows Yu et al. [5]; our negative DAPO result at n=5 paired Wilcoxon complements their positive single-seed results on math reasoning.

8 Limitations

(i) **Judge-sensitivity of the per-token KL headline.** On a n=72 stratified subset, the primary judge `gpt-5-mini` fires on 28% of iter 1 trajectories while the secondary judge `claude-haiku` fires on 6% (raw agreement 72.2%, Cohen's $\kappa = 0.08$). For comparison, the v4.1 baseline on the same subset has $\kappa = 0.42$ (primary 18% vs secondary 25%) — the judge agreement is *specifically poorer for the per-token KL student*, suggesting its outputs are at the judge threshold (borderline / context-sensitive failures that one judge calls harm and the other does not). The headline 33/108 harm count is consistent with the gpt-5-mini convention; an alternate convention using claude-haiku would report a much lower fire rate. We retain the gpt-5-mini headline because it was pre-registered, but the multi-judge sensitivity tempers interpretation of the absolute number; the *relative* comparisons (paired Wilcoxon vs v4.1) are by-construction judge-invariant. **Three-judge replication on a balanced subset** (GLM-4.6, n=60, 24 fires by primary, 49% fire rate after the harm audit) shows κ = 1.0 on `any_fire` across all three pairs (`gpt-5-mini`, `claude-haiku`, `claude-sonnet`; Fleiss $\kappa = 1.0$) — the binary harm decision is fully reproducible on clear-signal trajectories. The sub-flag κ collapses (≤ 0) for the rare `fabrication`/`deception` sub-flags (raw agreement $> 90\%$; this is the kappa paradox under low prevalence). The per-token KL judge gap therefore reflects *borderline-output ambiguity for the 8B student*, not a vendor-wide judge disagreement. (ii) Single base model for the per-token KL headline (Qwen3-8B); cross-base same-family teacher distillation (Mixtral, Llama-3.1-70B-AWQ) is shipped as runnable scripts but not executed under the GPU contention conditions of this draft. (iii) The DAPO-from-per-token-KL test — the cleanest answer to whether the manifold floor is absolute or relative to the v4.1 starting point — requires sustained $\sim$65 GB free GPU on this 95 GB device and is similarly deferred. (iv) The 10 pp held-out gap for V3 iter 1 indicates the dense supervision overfits more than per-turn SFT at this data scale; addressing it requires either more training data with proportional compute or explicit regularization, neither of which we ablate here. (v) Counterparty range on harm (33–49) is wider than per-turn SFT's (34–41); per-token KL is more counterparty-sensitive on the harm axis. (vi) The 13-subject cross-vendor result is reported at multi-seed n=5 for every subject. (vii) Mistral-Large held-out could not be collected because this OpenRouter account's provider allowlist excludes the only provider serving `mistralai/mistral-large-2411` (training-set numbers were collected before the routing change).

9 Conclusion

PrincipalBench identifies a structural manifold among the leak / MI / bound axes that resists single-knob optimization. Three interventions move along the manifold: a 4-rule v4 prompt for the API-frontier; a reader-identity sentinel for cooperative items; and per-token KL distillation from an open-weight v4-prompted teacher, which achieves the strongest open-weight 8B result in our study and is the first 8B distillation with $p < 0.05$ multi-seed harm improvement at both of its two stopping points.

Reproducibility. All code, items, and trained checkpoints at https://github.com/bojieli/principal-loyalty. The per-token KL pipeline is `scripts/pertoken_kl_collect.py` (vLLM `prompt_logprobs` collection), `scripts/train_pertoken_kl.py` (QLoRA top-K KL loss), and `scripts/run_pertoken_kl.sh` (end-to-end driver). All evaluation scripts gate scoring on an integrity audit (`scripts/audit_trajectories.py`); reported numbers use only audit-passing trajectories.

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